

Diagrams as Generative Models

Why the Brain Prefers Diagrams

We can now substantially strengthen our initial architectural claim from ??: formal commutative diagrams do not merely provide a convenient pedagogical representation illustrating abstract case structure—they mathematically constitute the exact native *computational format* through which biological cognitive agents actively maintain and query their internal generative models representing relational environmental structure.

This strong structural claim draws support from powerful converging empirical evidence:

1. **Computational advantage** (Larkin & Simon, [-@larkin1987diagram]): Diagrams enable search, recognition, and inference operations that are computationally prohibitive in sentential format. A commutative case diagram allows the agent to verify consistency (does the direct path equal the composed path?) by simple spatial inspection.
2. **Free ride inferences** (Shimojima, [-@shimojima1996reasoning]): Properties of the diagram that

P600 Signals and Garden-Path Reanalysis in Diagrammatic Models

The standard predictive processing framework—for which active inference operates as the most formally mathematically developed version—provides a remarkably natural, mechanistically precise account detailing exactly how biological agents actively deploy these diagrammatic representations cognitively during real-time processing:

1. **Top-down structural predictions:** The currently active internal case diagram continuously generates precise, high-precision predictions anticipating incoming expected sensory input (e.g., the system computationally predicts “a nominative-marked noun phrase must immediately appear because the parsed transitive verb structurally demands an active agent”).
2. **Bottom-up prediction errors:** Incoming sensory words that physically violate the model’s top-down diagrammatic predictions instantly generate massive, measurable prediction

Three Falsifiable ERP Predictions

The predictive processing account generates quantitative, falsifiable predictions about neural responses to case-marking violations. In the active inference framework, a case-assignment violation triggers a *prediction error* whose amplitude scales with the precision of the violated expectation—which is precisely the enriched hom-value of the violated morphism:

$$\text{PE}(f) \propto \pi_f \cdot |\mu_{\text{predicted}} - \mu_{\text{observed}}| \quad (1)$$

where $\pi_f = \mathcal{C}(A, B)$ is the enriched weight (precision) of the morphism $f : A \rightarrow B$ and μ are the expected vs. observed case features. This yields three concrete electrophysiological predictions:

1. **P600 amplitude scales with morphism weight:** A case violation on a high-weight morphism (NOM→ACC in a transitive clause, $w = 0.9$) should elicit a larger P600 than a violation on a low-weight morphism (NOM→INS in an experiencer construction, $w = 0.4$). The ratio of P600 amplitudes should approximate the ratio of enriched weights.

Integration: Five Pillars Become One Generative Model

The full picture emerges when we combine all five pillars within the active inference framework:

1. **Case categories** (??) provide the *objects and morphisms* of the generative model—the vocabulary of roles and relations
2. **Categorical grammar** (??) provides the *composition rules*—how roles combine to form structured derivations
3. **DisCoCat / DisCoCirc** (??, ??, ??) provides the *semantic functor* and discourse extension—mapping syntactic structure to distributional meaning
4. **Enriched structure** (??) provides the *precision parameters*—graded weights that control inference
5. **Topos-theoretic bridges** (??) provide *transfer theorems*—ensuring consistency across formalizations

The active inference agent uses this combined structure as a single, integrated generative model. Each scenario it encounters—a sentence heard, a scene observed, an action planned—is interpreted by instantiating a case diagram from structure (1),

804 Automated Tests Confirm the Formalism Is Executable

The framework developed in this paper is not merely theoretical—it is computationally verified through a comprehensive implementation and test suite that exercises every categorical construction discussed above.

System Architecture and Categorical Core

The categorical core (`CaseCategory`, `EnrichedCategory`, `AlignmentFunctor`, `NaturalTransformation`) is implemented in Python with set-based object tracking and list-based morphism storage, enforcing categorical axioms at construction time. Six additional modules extend the core: `FluidSFunctor` (context-dependent alignment parameterized by volition), `CaseDiagramBelief` and active inference computations (variational free energy, prediction error, belief updating), the `src/daif/ subpackage` (7 modules, 24 symbols—full distributional RL inference: push-forward returns, quantile TD, VMP, Bethe FE, EIG, ERP profiles, policy selection, and metrics), `CasePOVM` and quantum case assignment (POVM-based probability via ?? in ??). `DitransitiveSentence` (three-argument verb